Gain-Only Evaluation Hides Failures: A Benefit, Failure, Cost and Applicability Ledger for LLM Sentence Expansion in AAC

Anonymous submission

Abstract

Augmentative and alternative communication (AAC) apps increasingly let a large language model expand a few selected words into a sentence that the system may speak on the user's behalf, sometimes after review. Fixes for meaning errors in that sentence are usually reported by their targeted gain alone. We argue that gain-only evaluation hides failures, and report a *ledger*: benefit, offline failure indicators, cost and applicability together, against a cheap comparator and a kill criterion fixed in advance. Our lead intervention is a runtime guard that regenerates when the output contradicts a cue the user supplied. Frozen and rerun on 78 fresh speech-act item families, it raised joint pass from 78.79% to 88.62% (+9.83 points; deleting an unsupplied "please" alone gave +3.47) and passed every fixed transfer criterion. Yet a scored pass is not a usable sentence. In a pilot audit of 20 repaired and 20 control texts, raters judged repaired texts usable 58.8% of the time against 89.5%, a gap of -30.7 points [-54.7, -2.0]. The 14 raters are not independent and the controls had never failed, so item difficulty is confounded: the result is consistent with a usability cost, not an established one. On a second pool there was no gap (77.8% vs. 77.8%). A larger audit (120 new texts, 3 independent raters per text, plus original-failed and strip-only arms) with a decision rule frozen beforehand will decide. Two bounded secondary interventions, a menu and a user-side cue, show other failures. Beyond constructed items, the cue's gain replicates on all 6 dimensions of human-written pairs (exploratory; 6–7 writers per contrast), and people's own keywords drop negation 20.8% of the time, against 100.0% for a deliberately crude compressor. We study stored text, not people, and no AAC user took part.

1 Introduction

People who use AAC often compose by selecting a few words or symbols, because every selection costs time and effort (Valencia et al. 2023; Cai et al. 2024). LLM keyword expansion turns `want go party` into a full sentence (Shen et al. 2022; Cai et al. 2024), which a system may speak on the user's behalf, sometimes after the user reviews it (Hancock, Naaman, and Levy 2020; Valencia et al. 2020). The model chooses more than words: whether something happened or did not, whether the sentence asks or tells, and whether it says "please". Interventions can reduce wrong choices: a system can check its output against what the user supplied and try again (Madaan et al. 2023; Gou et al. 2024), offer suggestions spanning the readings the keywords leave open, or let the user add a cue such as `[NOT]` (Valencia et al. 2024; Mao et al. 2025; Xiang et al. 2025). Such interventions are usually reported by the gain on the endpoint they target. That hides what a fix leaves unfixed, what it changes that was already right, and whether the repaired text is something a person could use. Offered wording matters: suggested text shifts what people write and believe (Arnold, Chauncey, and Gajos 2020; Jakesch et al. 2023; Williams-Ceci et al. 2026).

Thesis and ledger. Gain-only evaluation hides failures, so we report a ledger. Beside each benefit it records *offline failure indicators* (stored-text events a gain-only report omits, such as residual failures, edits to already-correct outputs and readings a menu no longer offers), the cost, and where the intervention applies. None of the indicators is a measured harm to a person. Our running case is keyword-to-sentence expansion with four LLMs on minimal-pair item families, built in concurrent work that specifies meaning-preservation requirements and tests the scorer itself (Anonymous 2027).

Contributions. (1) A ledger and its design rules: channel-controlled attribution decides what an intervention may be credited with; paired stored configurations, a zero-call comparator and kill criteria fixed in advance decide whether it merits a human study (§5). (2) One lead intervention, a commitment guard, evaluated end to end: a frozen transfer that passes its pre-specified criteria, a modest increment over a free comparator, and a human usability audit whose pilot is consistent with a cost and whose decisive replication has a frozen decision rule (§6). (3) Two bounded secondary rows, a menu with collateral loss and a user-side cue whose preservation clause fails on fresh items, and a bridge to realistic input from human-written pairs and human-chosen keywords. We ran no study with AAC users; every result is a change in stored text-system behavior.

2 Related Work

LLMs in AAC. LLM expansion and suggestion systems for AAC include KWickChat (Shen et al. 2022) and SpeakFaster (Cai et al. 2024), and AAC users have described how

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

such systems help and hinder them (Valencia et al. 2023). Several add user-side control over intent (Valencia et al. 2024; Mao et al. 2025; Xiang et al. 2025; Lacosse, Tano, and McNamee 2026); others measure whether affect or intent survives generation (Choudhury, Kumar, and Martin 2025; Gorenshtein et al. 2026). Frisch et al. argue that AI-powered AAC needs technical, user-mediated and functional evaluation together (Frisch et al. 2026); our ledger is a technical layer shaped so that a functional study can test it.

Editing, preservation and evaluation. Generate–critique–revise loops (Madaan et al. 2023; Gou et al. 2024) and conditioning a rewrite on the source dialog act (Qu, Ungar, and Sedoc 2023) are established, as is reporting a targeted change together with collateral change (Ziegenbein, Stahl, and Wachsmuth 2026; Elgaar and Amiri 2025). Style-transfer meta-evaluation shows that preservation metrics look valid only when test data do not confound the intended change with content errors (Pauli, Augenstein, and Assent 2025); detoxifying rewriters default to polite output that distorts intent (Wang et al. 2025), and models have default politeness strategies (Zhao and Hawkins 2025). We claim neither the loop nor gain–harm accounting. We add accounting conditioned on what the input *supplied*, against a zero-call comparator, plus a human usability check on repaired text. Opinionated co-writers shift users' views (Jakesch et al. 2023), AI replies change language and social perception (Hohenstein et al. 2023), and suggestions pull writing toward dominant styles (Agarwal, Naaman, and Vashistha 2025); those are human-outcome studies of exposure, whereas ours is a pre-deployment study of interventions meant to *prevent* unwanted shifts. LLM judges (Zheng et al. 2023) can agree with each other more than with people on pragmatic labels (Wang, Sun, and Guo 2026; Panickssery, Bowman, and Feng 2024), which bounds what our machine labels can show.

3 Setting

Items and populations. Items are minimal-pair families ("I don't like the new medication" / "I like the new medication") rendered through five channels (Table 1). We use populations from the concurrent study (Anonymous 2027). The *discovery benchmark* has 58 held-out families on six dimensions, after quarantining two families whose scoring rule has a known blind spot; the guard is developed and tested there, and all I2 results use it. *W-C* has 240 fresh families (polarity, speech act, politeness), 238 analyzed after a pre-generation exclusion, with three contrasts pre-specified and git-timestamped; its confirmation is qualified because 48 cached requests (12 distinct primary ones) predate the final item freeze. Its 78 speech-act families are the separate population for the frozen guard transfer. *Human-written pairs (hw1)* are 547 kept minimal pairs written by non-author annotators, and *elicitation* has people type their own keywords for 600 crowd-written AACText messages (fictional messages written by crowd workers imagining themselves as AAC users, not messages of AAC users) (Anonymous 2027). Families are otherwise constructed. The AAC input studied is word input only; symbol and grid-based AAC are out of scope.

Attribution decides what an intervention may claim. Because the channel fixes which commitments the model received, every miss is attributed by construction (Anonymous 2027). A checker can only enforce a commitment the user supplied, so the guard is credited with Violations alone. When the keywords leave a dimension open, no checker can know the intended reading; the remedies are to show alternatives (I2) or to let the user supply the cue (chip).

Systems and scoring. Systems are Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-4B-it (4-bit, llama.cpp) and the API model gpt-6.1-terra. An LLM extractor (gpt-6.1-sol, two prompt variants) labels each output's dimensions without seeing source, model or arm; a cross-family arbiter resolves critical-label disagreements, and deterministic code computes verdicts, identically for treated and untreated outputs. In W-C, 259 disagreement rows (177 units) were resolved by a single arbiter after outcomes were known, blind to reference, model and arm; masking them as unknown gives the same three conclusions. Against two human annotators on 163 sampled untreated outputs, extractor labels agree at held-out $\kappa = 0.91$–0.97 on all 7 label fields (pre-specified gate 0.6). That gate covers untreated outputs; it does not validate repaired or rewritten text.

4 Interventions

I1, the commitment guard (lead; system side). After the first-pass P0 output, a deterministic lexical monitor (regexes and lexicons for negation, person, modality, question form, numbers, tense and "please") compares the output with the commitments in the visible input: the chips, or for full input the monitor's own labels of the typed sentence. It also flags a politeness marker that was not supplied. On a flag, the same conversation receives one feedback turn ("Your sentence must keep: negated; no added politeness. Write the sentence again with the same meaning otherwise.") and is regenerated, up to two retries. The guard returns the first unflagged attempt, else the attempt with fewest flags, ties going to the original. The strip-only comparator G1s deletes an unsupplied "please", "pls" or "plz" at zero calls.

I2, ambiguity-complete suggestion set (secondary; bare input). Starting from the P2 set, I2 keeps the top suggestion and, for polarity then certainty, rewrites the third (then second) slot into a missing reading with one LLM call, verified by the monitor with one feedback retry. It was chosen over a variant on a 12-family development pilot before the test run.

Chip (secondary; user side). The user adds one commitment cue, such as `[NOT]`, `[MAYBE]`, `[?]` or `[PLAIN]`. We treat it as a *probe* of whether information the user supplies is used, not as a proposed interface. Its real cost depends on the access method (one dwell in eye gaze, a menu traversal in scanning, a fixed button in a symbol grid), and it adds a cue vocabulary to learn, a decision for every message, and a risk that cues are copied literally into the output; none of these is measured here.

Glossary of terms and channels			
Family	Two source sentences differing on one critical dimension, with the same bare keywords	*bare*	Keywords only (`like medication`)
Critical dimension	Polarity, agent, certainty, speech act, politeness, time/quantity	*noncrit*	Keywords plus one non-critical word (length control)
Loss	The channel never carried the value	*chip1*	Keywords plus the critical cue (`[NOT] like medication`)
Violation	The channel carried it and the output changed it	*chipall*	Cues for all commitments
Assumption	The channel was ambiguous and the output committed anyway	*full*	The typed source sentence
Strict pass	Keeps the critical value, contradicts no supplied dimension, adds no politeness	*P0 / P2 / S3*	One sentence (greedy) / three suggestions covering different meanings / three samples at temperature 0.7
Joint pass	Keeps the critical value and every other scored supplied predicate	*G1 / G1s*	Guard output / strip-only comparator (deletes an unsupplied "please")
Offline failure indicator	Stored-text event a gain-only report omits	*Chip*	A probe: one user-added commitment cue

Table 1: Glossary. All scores are partial predicates computed from machine-extracted labels, not judgments of usability.

5 Evaluation Design

Paired comparisons of stored configurations. For system-side interventions, the untreated outcome is the *stored* first-pass output for the same request, and the treated outcome is what the intervention returns from it. This removes between-request variation but does not identify an effect on people. Pairing matters because greedy decoding is not reproducible: issuing the same request twice reproduced the text in only 76.4–90.8% of local and 44.4–71.4% of API requests. Estimands are row-weighted means over targets; the four models are fixed, not sampled. Intervals are 95% percentile intervals from whole-family or cluster resamples (2,000 on discovery, 10,000 elsewhere; fixed seeds), are pointwise, and do not correct for exploratory selection or label error. Thousands of rows are not thousands of independent tests: the guard's 63/80 discovery repairs come from 26 families.

Offline failure indicators and cost. *Changed-correct* counts already-correct outputs whose text the intervention changed; *correct→failure* is the share that became scored failures; *residual* failures remain after the intervention. We also show what a label cannot see, such as scored passes that are fragments, and, for repaired text, a human usability rating. Cost is extra generator calls and latency (local latency is not a deployment prediction). Kill rules are coarse by design: they decide whether an intervention merits a human study, not whether it is good.

6 Results

6.1 The guard: most scored failures repaired, a modest increment over a free comparator

On discovery cue channels, strict pass rose from 94.25% to 98.78% of 1,392 targets (Table 3). Of the 80 initial failures the guard repaired 63 (78.8% [68.4, 88.9]), far above its 30% kill line. A gain-only report stops there. Deleting an unsupplied "please" at zero calls already reaches 1,346 passes, so regeneration's increment is 2.08 points [1.15, 3.16]: 31 targets pass only with the guard and 2 only with deletion. This compares stored configurations and does not isolate the feedback message. 17 of 80 failures remained, 12 of them involving speech act, and 14/17 were still flagged after both retries: detected, not repaired. The guard changed the text of 49/2,154 already-correct outputs, none into a scored failure; yet "Would you mind repeating the last question" became "Please repeat the last question", still polite by the label and no longer what was typed. It flagged 157/1,856 distinct inputs and added 0.104 generator calls per input.

Frozen transfer. Speech act was the guard's weakest dimension, so we reran the guard unchanged on W-C's speech-act families (78 families, four models, three cue channels, 1,872 targets). Criteria were frozen before any retry: at least 30% of strict failures repaired, at most 10% of strict-correct texts changed, and a joint-pass gain whose lower 95% bound exceeds zero. Joint pass rose from 78.79% to 88.62%, 9.83 points [8.39, 11.32]; the guard repaired 251/478 strict failures (52.51% [48.36, 56.70]), changed 26/1,394 strict-correct texts and turned 0 into scored failures. Every transfer criterion passed within the call caps. Deletion alone gained 3.47 points, so the increment over the comparator is 6.36 points [5.29, 7.48]. The residue is larger too: 12.13% of targets remain strict failures. Gains concentrate where first passes were worst (19.02 points for Gemma and 16.24 for Llama, against 3.21 for Qwen and 0.85 for Terra). The declared changed-correct example shows the gap: "Is it okay if I bring a friend?" became "Is [I] bring a friend?", which keeps the question label and so remains a scored pass.

6.2 What the pass does not show: repair usability

A pass checks a few labels, not a sentence. For `[ASK] [YOU] [PLAIN] wear headphones`, the guard turns "You wear headphones." into "Ask you to wear headphones.", which passes every scored predicate; 4/63 scored repairs begin with this chip echo, a post hoc lexical count. We therefore had people judge repaired text. In a pilot audit, 14 reim-

	I1 guard (lead)	I2 set (bounded)	Chip probe (bounded)
Defect targeted	Violation of a supplied commitment	Assumption under ambiguous input	Loss of a cue at input
Comparison	Stored P0 first pass, same request; strip-only G1s (0 calls)	Stored P2 set, same request	Length-matched non-critical word
Population	Discovery: 1,392 cue targets; frozen transfer: 1,872 W-C speech-act targets	Discovery bare-input sets of three	Discovery: 464 targets; W-C speech act: 624 targets
Primary benefit	Strict pass; joint pass on transfer	Sets offering both readings	Critical-value match
Offline failure indicators	Residual failures; correct text changed; scored passes that are fragments; *human-rated usability of repairs*	Coverage lost on other dimensions; rewrites changing sentence type	Matches lost; misses unrepaired; gap to full input
Applicability	Supplied cues only	Ambiguous input only	Dimension-specific
Cost	Generator calls, latency	Generator calls, latency	Selections (access-dependent)
Kill criterion	Repair $\geq$30% of failures *and* change $\leq$10% of correct outputs; transfer joint-pass gain with lower bound > 0; deployment clause (usable $\geq$ control -10 points) frozen for Part 17	Gain $\geq$30 points over P2	Gain $\geq$10 points *and* $\geq$80% of full-input match

Table 2: Evaluation card: intervention, comparison, population, outcomes. Verdicts are in Table 3, Fig. 1 and §6.2. Discovery kill criteria were fixed in the analysis code and checked on development data before the held-out run, without external timestamps; the transfer criteria were frozen in a hashed file before any retry request.

bursed non-author raters judged 20 repaired outputs and 20 unchanged controls (outputs that had never failed) for usability, blind to arm. By item consensus, repaired texts were usable in 10/17 rated items (58.8%) against 17/19 (89.5%) for controls, a gap of -30.7 points [-54.7, -2.0], 3 and 1 items being unresolved. The raters are not independent, the controls had never failed so repaired and control items differ in difficulty, and the interval is wide: the pilot is *consistent with a usability cost*; it does not establish one. On the second pool of the same design (transfer outputs) the gap was absent: 77.8% against 77.8%. Table 4 shows what unusable repairs look like: chip echoes, dropped content words and fragments, all passing the scored predicates. *Part 17* decides the question: 120 new texts, 3 independent raters per text, with original-failed and strip-only arms to separate repair from difficulty. Its decision rule was frozen before any return (pooled interval entirely below -10 points: killed for deployment, kept as a detector; including -10 but excluding 0: a cost of uncertain size; including 0: no detectable cost), and neither audit is selected for being favorable.

6.3 Secondary rows: the menu and the cue probe

I2 (bounded). Under bare input, P2 menus offered both polarity or both certainty readings in 15/80 sets; I2 raised this to 78/80, a paired gain of 78.75 points [70.00, 87.50], passing its kill line. On separate politeness families, sets offering both a plain and a polite reading fell from 26/40 to 12/40 (-35.00 points), and on speech-act families from 14/31 to 6/31 (-25.81 points); the gains and losses are on different strata, not a within-item trade. For `knock come`, "Please knock before you come in." became "Don't knock before you come in.", removing the polite option. Of 428 rewrites, 138 (32.2%) changed sentence type. Applicability matters: when the input fixes a value, a menu prompt asking for different meanings departs from it, with a focal-label departure in 62.76% of P2 menus against 2.70% of sample sets for typed input. That gap reflects prompt design (one prompt asks for different readings, the other draws samples), and does not separate prompt from temperature; it supports conditioning diversification on which dimensions are open.

Chip probe (bounded). On discovery, one chip raised critical-value match from 41.8% with the length-matched word to 94.8% (97.3% of full input), passing both kill clauses. On 78 fresh W-C speech-act families, a contrast pre-specified and git-timestamped, it raised preservation from 41.83% to 73.24%, 31.41 points [25.00, 37.94], so the gain clause replicates. The $\geq$80% clause does *not*: full input preserved speech act in 99.20%, so one chip reaches 73.8% of it. The chip therefore passed its discovery test and failed one clause of its fresh-item test. Question cues fail most visibly: 58/208 question targets became non-questions, as when `[?] parcel signature` became "Can you sign for this parcel?". A second cue is itself an input intervention: Llama under one encoding lost an already-preserved negation in 55/640 transitions where `[NOT]` meets `[YOU]` (other cells 0/480), which supports per-model, per-encoding regression tests for cue combinations.

6.4 Bridge to realistic input

Human-written pairs (hw1). Constructed families may not look like what people write, so non-author annotators wrote 547 kept minimal pairs, run through the same pipeline with the same four models. The chip gain over the length-matched word replicates on all 6 dimensions (35.3 to 82.7 points); the three W-C contrasts replicate in direction: speech-act chip +35.3 points [31.2, 40.3] (W-C +31.41), politeness addition +7.6 [5.0, 10.4] (W-C +6.25), and a bare affirmative default 99.8% [99.3, 100.0]. This replication is exploratory and not pre-specified, each contrast rests on only 6–7 writers (so we claim direction and consistency, not interval width), and on binary dimensions the design caps the

	Discovery 58 families, 6 dimensions, 1,392 cue targets	**W-C transfer (frozen)** 78 families, speech act, 1,872 targets
Benefit (% of targets)		
Strict pass P0 / G1s / G1	94.25 / 96.70 / 98.78	74.47 / 81.73 / 87.87
Joint pass P0 / G1s / G1	94.68 / 96.77 / 98.85	78.79 / 82.26 / 88.62
G1−P0, points	+4.53 strict	**+9.83** [8.39, 11.32] joint
Strict failures repaired	63/80 (78.8%)	251/478 (52.51%)
Comparator: strip-only G1s, zero calls		
G1s−P0, points	+2.44 strict	+3.47 [2.78, 4.22] joint
G1−G1s, points	**+2.08** [1.15, 3.16] strict	+6.36 [5.29, 7.48] joint
Passes only G1 / only G1s	31 / 2	123 / 4
What remains (offline failure indicators)		
Residual strict failures	17/1,392 (1.2%)	227/1,872 (12.13%)
Correct text changed	14/1,312 cue; 49/2,154 all ch.	26/1,394 (1.87%)
Correct → failure	0/1,312	0/1,394
Cost and verdict		
Extra calls per input	0.104 (1,856 inputs)	0.41 (1,848 inputs)
Kill / transfer rule	passed	passed (all criteria)

Visible input	P0 output	G1 output
Discovery, repaired (Gemma, pol2-14/A, chip1; 1st by hash order; 63/80 scored failures repaired)		
`[PLAIN] let finish sentence`	"Please let me finish my sentence."	"Let me finish the sentence."
Discovery, scored repair that is a fragment (Llama, pol2-07/B, chipall; 2nd by hash order; 4/63 scored repairs begin "Ask you", a post hoc lexical count, not a fluency rate)		
`[ASK] [YOU] [PLAIN] wear headphones`	"You wear headphones."	"Ask you to wear headphones."
Discovery, residual (Llama, act-11/A, chipall; 1st by hash order; 17/80 failures remain, 14/17 still flagged; the source asks whether the listener is charging it)		
`[?] [YOU] charge wheelchair`	"Can you charge my wheelchair?"	"Can you charge my wheelchair?"
W-C transfer, correct text changed, still a scored pass (Qwen, r2act-073/A, chipall; 1st by SHA-256 order; 26/1,394 strict-correct targets changed, none to a scored failure)		
`[?] [I] bring friend`	"Is it okay if I bring a friend?"	"Is [I] bring a friend?"

Table 3: The guard as one display. Left: benefit, the zero-call comparator and what remains, on two *separate* populations that are never pooled; scored partial predicates, not usability. Right: stored examples, verbatim, each following the declared outcome-conditioned rule named in its row; they are not typical. Brackets: 95% whole-family bootstrap intervals for paired differences. Repair rates condition on initial strict failures; changed-correct and correct→failure on initial strict passes.

Pool	Visible input	Stored repaired output
Pilot	`[ASK] [YOU] [PLAIN] speak slowly`	"Speak slowly to you."
	`[ASK] [YOU] [PLAIN] charge wheelchair`	"Ask you for the wheelchair charge."
	`evening take pills`	"Take pills evening."
Transfer	`[ASK] [YOU] [PLAIN] email manager`	"Ask you email manager."
	`[ASK] turn conditioner`	"Give me a turn conditioner."
	`[TELL] text plumber`	"The plumber."

Table 4: Repairs raters judged unusable by consensus, verbatim. Rule: the first three consensus-unusable repaired items in frozen hash order in each pool (pilot: 7/17 of rated repairs unusable; transfer: 4/18). Chip echoes ("Ask you", "Tell you") begin 3 of the three pilot and 1 of the three transfer examples.

gain near 50 points; sensitivity checks moved no estimate by more than 4.4 points.

Human-chosen keywords. In the full elicitation, 20 annotators typed keywords for 600 crowd-written AACText messages (1800 sets). People's keywords lost the negation cue in 20.8% of sets [16.9, 25.2], against 100.0% for a rule-based four-word compressor. That comparison is against a deliberately crude comparator and shows only that human keywording is not that compressor; it does not say how often real AAC users drop negation. Politeness reverses: people lost the polite cue more often (35.7%) than the compressor (17.3%). People also disagree with each other on what their keywords carry (Krippendorff $\alpha = 0.61$).

7 From Ledger to Testable Predictions

The ledger is a pre-deployment screen, useful only if it predicts human validity. We therefore state falsifiable predictions, each with the result that would demote it. **H1 (writer intent).** Among outputs the scorer counts as preserving a writer's critical value, writers report that the output "said what I meant" more often than among outputs it counts as failing; a difference whose interval includes zero demotes the scored endpoint. *Part 18*, the first test, has each writer judge 10 outputs from their own kept pairs (bare and chip1);

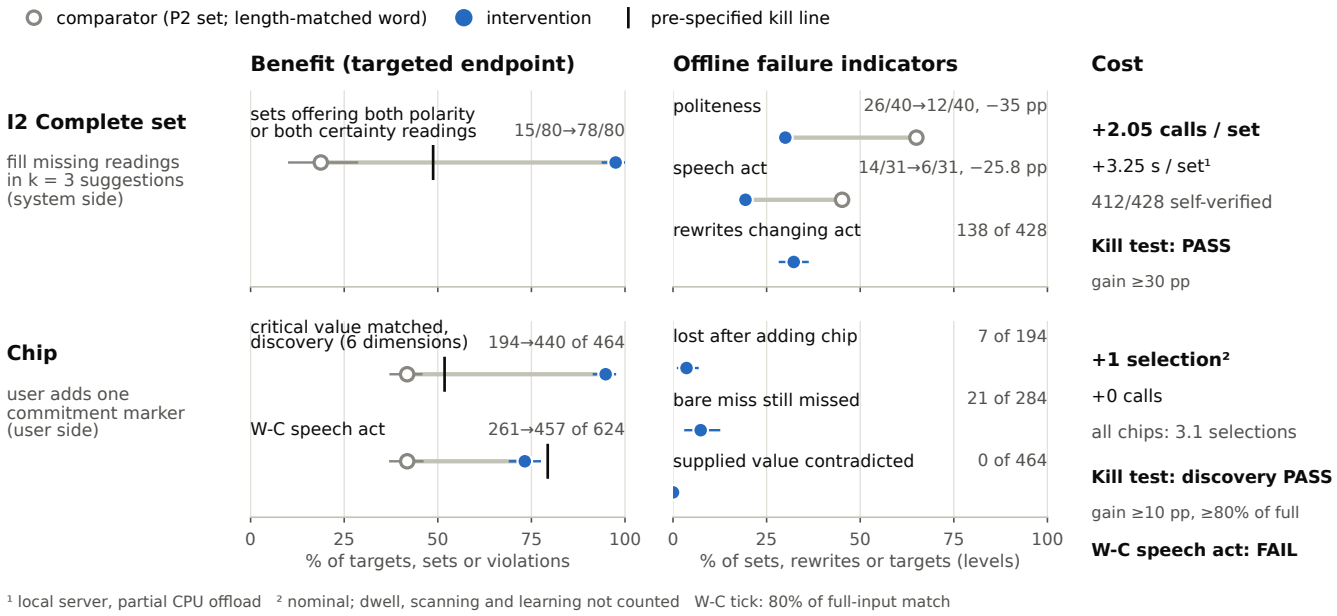


Figure 1: Ledger rows for the two bounded secondary interventions (machine-extracted labels; four models). Open circle: comparator; filled: intervention; bars: 95% cluster-bootstrap intervals; black tick: the pre-specified kill line. The chip passes both clauses on discovery but *fails* its ≥80%-of-full clause on fresh W-C speech-act items (only its gain clause replicates). Plotted values are percent *levels*; changes in points are printed. I2's politeness and speech-act rows come from families different from its target gain (40 and 31 sets). The guard is in Table 3.

it measures writer intent, never AAC-user validation. **H2 (repair usability).** A scored repair is usable at no less than 10 points below a never-failed control; Part 17 tests this and its frozen rule (§6.2) already names the outcome that retires the guard as a deployment feature. **H3 (collateral).** Offline changed-correct and coverage loss predict a later drop in the user's "said what I meant" rate; if not, evaluation of this kind should be demoted. H3 requires design with AAC users: a micro-randomized trial (Klasnja et al. 2015) over weeks of users' own use, randomizing only where the ledger shows arms differ (silent regeneration vs. flag-and-confirm vs. original; I2 vs. P2 menus), estimating causal excursion effects (Boruvka et al. 2018) on proximal outcomes such as sent without edit. Chips could only be offered, an encouragement design with its own estimand. This is a proposal, unrun, and the guard's detection value suggests flag-and-ask rather than silent rewrite as an arm.

8 Limitations

No human outcomes. Nothing here measures communication success, agency, effort or trust; offline failure indicators are not experienced harms. *Repair audit.* The pilot has 20 + 20 items and non-independent raters, and its controls never failed; Part 17 is the primary estimate. *Machine labels.* Endpoints use an LLM extractor and deterministic verdicts, validated by two human annotators on untreated outputs only; guard repairs, I2 rewrites and new boundary cases are not covered by that gate, and shared monitor and extractor errors can inflate repairs. *Scope.* The guard has two separate populations; I2 was run on neither W-C nor supplied input; W-C's confirmation is qualified by early cached requests and post-outcome arbitration; the applicability contrast and W-D are exploratory; hw1 is exploratory with few writers. *Constructed items, fixed models.* Four fixed models describe these systems, not other models; AACText is crowd-written fiction, and word input is the only modality studied. *Authorship.* Many AAC users compose by rewriting or choosing among options rather than typing keywords, and we do not model this. *Selection.* I2 was chosen over a variant on development data; examples follow declared, outcome-conditioned rules and illustrate rather than sample. *Community involvement.* No AAC user or autistic adviser contributed to the design or analysis; we draw on guidance such as AASPIRE (Nicolaidis et al. 2019) and on the double-empathy (Milton 2012) and heterogeneity (Zisk and Dalton 2019) cautions, assume no "autistic style", and limit claims to commitments explicit in the text. Any claim about benefit requires direct involvement of AAC users and advisers. *Ethics.* No data were collected from AAC users. All annotators were adult non-authors recruited from the authors' university peers, identified only by an ID; each consented and was reimbursed for their time (payment and food). They labelled model outputs and audited items under a fixed codebook, blinded to model, arm and source; some also chose keywords for public AACText messages or wrote new sentence pairs, released under CC BY 4.0 with recorded consent. No personal or health data were collected, and we did not seek an institutional review determination.

9 Conclusion

Gain-only reporting would have credited the guard with its transfer gain and stopped. The ledger adds that deletion alone explains much of it, that scored passes can be fragments, and that a pilot audit is consistent with a usability cost that a frozen, larger audit will settle. The two bounded rows show the same pattern: the menu gains target readings and loses others, and the chip passed its discovery test but failed its $\geq$80%-of-full clause on fresh items, so not every intervention passed every pre-specified test. Human-written pairs and human-chosen keywords suggest the effects are not artifacts of authored items, but only people who use AAC can say whether any of it matters to them. We release the analysis code, stored outputs and scripts that rebuild every number and figure.[PENDING: anonymized artifact URL]

References

Agarwal, D.; Naaman, M.; and Vashistha, A. 2025. AI Suggestions Homogenize Writing Toward Western Styles and Diminish Cultural Nuances. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM.

Anonymous. 2027. Testing What an AAC Expander Is Allowed to Say: Input-Conditioned Commitment Contracts for LLM Sentence Expansion. Concurrent submission to SE4All at ICSE 2027; anonymized copy provided to reviewers.

Arnold, K. C.; Chauncey, K.; and Gajos, K. Z. 2020. Predictive Text Encourages Predictable Writing. In *Proceedings of the 25th International Conference on Intelligent User Interfaces*. ACM.

Boruvka, A.; Almirall, D.; Witkiewitz, K.; and Murphy, S. A. 2018. Assessing Time-Varying Causal Effect Moderation in Mobile Health. *Journal of the American Statistical Association*, 113(523): 1112–1121.

Cai, S.; Venugopalan, S.; Seaver, K.; Xiao, X.; Tomanek, K.; Morris, M. R.; Kane, S.; Narayanan, A.; MacDonald, R. L.; Kornman, E.; Vance, D.; Casey, B.; Gleason, S. M.; Nelson, P. Q.; and Brenner, M. P. 2024. Using large language models to accelerate communication for eye gaze typing users with ALS. *Nature Communications*, 15: 9449.

Choudhury, S. H.; Kumar, A.; and Martin, L. J. 2025. Evaluating Human-LLM Representation Alignment: A Case Study on Affective Sentence Generation for Augmentative and Alternative Communication. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, 1618–1637. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics.

Elgaar, M.; and Amiri, H. 2025. Linguistically-Controlled Paraphrase Generation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 20842–20864. Association for Computational Linguistics.

Frisch, B.; Wade, W.; Gaines, D.; Kinsella, M.; Peters, B.; Broderick, T.; and Vertanen, K. 2026. It's Complicated: On the Design and Evaluation of AI-Powered AAC Interfaces. Position paper, Speech AI for All workshop at CHI 2026, arXiv:2606.24854.

Gorenshtein, A.; Omar, M.; Jia, E.; Adiniaev, Y.; Daniel, O.; Kruskal, J.; Ahmed, M.; Brook, O.; Klang, E.; and Barash, Y. 2026. Intent Drift in LLM-Assisted BCI Communication: An In-Silico Benchmark Under Simulated Decoder Corruption. MedRxiv preprint, posted 27 August 2026.

Gou, Z.; Shao, Z.; Gong, Y.; Shen, Y.; Yang, Y.; Duan, N.; and Chen, W. 2024. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. In *International Conference on Learning Representations*.

Hancock, J. T.; Naaman, M.; and Levy, K. 2020. AI-Mediated Communication: Definition, Research Agenda, and Ethical Considerations. *Journal of Computer-Mediated Communication*, 25(1): 89–100.

Hohenstein, J.; Kizilcec, R. F.; DiFranzo, D.; Aghajari, Z.; Mieczkowski, H.; Levy, K.; Naaman, M.; Hancock, J.; and Jung, M. F. 2023. Artificial intelligence in communication impacts language and social relationships. *Scientific Reports*, 13: 5487.

Jakesch, M.; Bhat, A.; Buschek, D.; Zalmanson, L.; and Naaman, M. 2023. Co-Writing with Opinionated Language Models Affects Users' Views. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM.

Klasnja, P.; Hekler, E. B.; Shiffman, S.; Boruvka, A.; Almirall, D.; Tewari, A.; and Murphy, S. A. 2015. Microrandomized Trials: An Experimental Design for Developing Just-in-Time Adaptive Interventions. *Health Psychology*, 34(Suppl.): 1220–1228.

Lacosse, E.; Tano, P.; and McNamee, D. C. 2026. Multi-System Cognitive Steering for Generative Communication. Preprint, version 1, posted 13 October 2026; manuscript identifies the BiAlign Workshop at CHI 2026, arXiv:2610.19177.

Madaan, A.; Tandon, N.; Gupta, P.; Hallinan, S.; Gao, L.; Wiegreffe, S.; Alon, U.; Dziri, N.; Prabhumoye, S.; Yang, Y.; Gupta, S.; Majumder, B. P.; Hermann, K.; Welleck, S.; Yazdanbakhsh, A.; and Clark, P. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems*, volume 36.

Mao, L.; Lee, J. H.; Faroqi-Shah, Y.; and Valencia, S. 2025. Design Probes for AI-Driven AAC: Addressing Complex Communication Needs in Aphasia. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*, 404–422. ACM.

Milton, D. E. M. 2012. On the Ontological Status of Autism: The 'Double Empathy Problem'. *Disability & Society*, 27(6): 883–887.

Nicolaidis, C.; Raymaker, D.; Kapp, S. K.; Baggs, A.; Ashkenazy, E.; McDonald, K.; Weiner, M.; Maslak, J.; Hunter, M.; and Joyce, A. 2019. The AASPIRE Practice-Based Guidelines for the Inclusion of Autistic Adults in Research as Co-Researchers and Study Participants. *Autism*, 23(8): 2007–2019.

Panickssery, A.; Bowman, S. R.; and Feng, S. 2024. LLM Evaluators Recognize and Favor Their Own Generations. In *Advances in Neural Information Processing Systems*, volume 37.

Pauli, A. B.; Augenstein, I.; and Assent, I. 2025. Mind the Style Gap: Meta-Evaluation of Style and Attribute Transfer Metrics. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 21550–21564. Association for Computational Linguistics.

Qu, R.; Ungar, L.; and Sedoc, J. 2023. Conditioning on Dialog Acts improves Empathy Style Transfer. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 13254–13271. Association for Computational Linguistics.

Shen, J.; Yang, B.; Dudley, J.; and Kristensson, P. O. 2022. KWickChat: A Multi-Turn Dialogue System for AAC Using Context-Aware Sentence Generation by Bag-of-Keywords. In *Proceedings of the 27th International Conference on Intelligent User Interfaces*, 853–867. ACM.

Valencia, S.; Cave, R.; Kallarackal, K.; Seaver, K.; Terry, M.; and Kane, S. K. 2023. "The less I type, the better": How AI Language Models can Enhance or Impede Communication for AAC Users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM.

Valencia, S.; Huynh, J.; Jiang, E. Y.; Wu, Y.; Wan, T.; Zheng, Z.; Admoni, H.; Bigham, J. P.; and Pavel, A. 2024. COMPA: Using Conversation Context to Achieve Common Ground in AAC. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. ACM.

Valencia, S.; Pavel, A.; Santa Maria, J.; Yu, S.; Bigham, J. P.; and Admoni, H. 2020. Conversational Agency in Augmentative and Alternative Communication. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM.

Wang, R.; Sun, K.; and Guo, Y. 2026. Polite but Misaligned: Evaluating LLM Politeness Judgments Against Human Pragmatic Norms. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, 26402–26421. Association for Computational Linguistics.

Wang, X.; Liu, Y.; Pan, J.; Ding, L.; Wang, L.; and Biemann, C. 2025. Chinese Toxic Language Mitigation via Sentiment Polarity Consistent Rewrites. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 35695–35711. Association for Computational Linguistics.

Williams-Ceci, S.; Jakesch, M.; Bhat, A.; Kadoma, K.; Zalmanson, L.; and Naaman, M. 2026. Biased AI writing assistants shift users' attitudes on societal issues. *Science Advances*, 12(11): eadw5578.

Xiang, W.; Xu, Y.; Fang, Y.; Teng, Z.; Jiang, Z.; Hu, B.; and Yang, J. 2025. SocializeChat: A GPT-Based AAC Tool Grounded in Personal Memories to Support Social Communication. Accepted at IEEE SMC 2025 per author record, arXiv:2510.19017.

Zhao, H.; and Hawkins, R. D. 2025. Comparing human and LLM politeness strategies in free production. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 16188–16216. Association for Computational Linguistics.

Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E. P.; Zhang, H.; Gonzalez, J. E.; and Stoica, I. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36.

Ziegenbein, T.; Stahl, M.; and Wachsmuth, H. 2026. Teaching LLMs Human-Like Editing of Inappropriate Argumentation via Reinforcement Learning. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 38616–38637. Association for Computational Linguistics.

Zisk, A. H.; and Dalton, E. 2019. Augmentative and Alternative Communication for Speaking Autistic Adults: Overview and Recommendations. *Autism in Adulthood*, 1(2): 93–100.